

Tracking the Trackers

An Observation and Analysis of an Information-Seeking Event on Digital Privacy

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Introduction

The information-seeking event observed for this analysis was a sixty-minute self-directed session aimed at developing a working conceptual understanding of two related domains: how personal data is collected as a user moves across the web, and how the most common technical privacy mechanisms operate at a functional level. The session took place using Mozilla Firefox as the primary browser, DuckDuckGo as the initial search engine (consistent with the topic's framing), and with standard privacy extensions active: uBlock Origin, Privacy Badger, and Multi-Account Containers. A simultaneous screen recording with a think-aloud commentary was made and later transcribed; the edited transcript constitutes the primary raw data for this analysis and is edited for clarity in the Appendix.

This need is organic in origin. No course assignment required that this search happen on this day, nor did any external deadline sharpen its urgency. Instead, the motivation had been quietly accumulating for some time, intensified by two convergent pressures: coursework in open-source intelligence (OSINT) that made the aggregation of personal information feel concrete rather than abstract, and the broader

cultural normalization of large language models whose training pipelines routinely harvest publicly available web content. The combination produced what Taylor (1968) described as a conscious level need: the seeker could name the general problem area and articulate a rough sense of what was missing, but could not yet formalize it into a precise, answerable question. The vocabulary to do so had not yet been acquired. Developing that vocabulary, and with it a conceptual map of the domain, was the true goal of the session.

The subjective importance of this need was high, higher than the seeker's pre-session framing as "moderate-to-high" initially acknowledged. What the session produced was not mere cautious intellectual curiosity about a separable external subject but a felt sense of exposure tied to information infrastructure the seeker was already inside every day. Unlike a lookup query with low personal stakes, this event involved information about the infrastructure through which all other information seeking takes place. To seek information about tracking is to try to understand the very system one is already embedded in, which gives the need a distinctive reflexive quality that most everyday information-seeking events do not share. Experiencing that sense of exposure that had built over time without producing action, is a pattern consistent with Bawden and Robinson's (2009) account of information anxiety: the knowledge that a gap exists, combined with uncertainty about how large it is and what filling it would require, had itself become a source of paralysis.

Description of the Event

The session opened not with the broad query "how does browser tracking work" but with something more revealing of the seeker's actual starting state: "what does a website learn about me when I visit." Early results included the FTC's consumer explainer, quickly dismissed as "more occupied with actionable recommendations rather than actually detailed explanations," and [privacy.net](https://www.privacy.net), which was opened but deferred. The session gained its first real traction at givemeyourdata.org, a demonstration tool that displayed the seeker's device data in real time: browser fingerprint, IP address, referrer information. The seeker's reaction was immediate, "Wow they just straight show me all the details available on my device," and included the observation that the site resembled what they eventually hoped to build: an accessible, user-oriented information security guide.

The site's URL tracking demonstration, however, produced the session's first significant friction. Invited to add parameters to a URL to observe tracking, the seeker could not identify any change. The annotation in the transcript captures this saying, "when demonstrating something invisible, name the invisibility. Don't let the user think the demo doesn't work." This failure became the session's first design principle, generated inductively from the seeking process itself.

The bulk of the session's middle section was structured around cookies and pixels, not the VPN/DNS material that the seeker might later nominate as most intellectually significant. The seeker moved through UTM parameters, gclid/fbclid identifiers, a Usercentrics article on tracking pixels, and the learnwebanalytics comparison diagram, before arriving at minute 11:55 at MDN's documentation on HTTP cookies. It was here, not at the DNS moment, that the first formulation occurred. Reading that the browser checks stored cookies and includes matching ones in every HTTP request automatically, the seeker wrote out the mechanism in their own words:

"When I land on a 'Thank You' page that has an advertising pixel on it, my browser fetches the pixel from the ad server. That fetch is a request. Because I've previously visited some other site that uses the same ad server (via third-party cookies), my browser already has a cookie for the ad server's domain. So the pixel-fetch request automatically carries that cookie. The ad server now knows: 'the same browser that I previously identified by this cookie is now on this Thank You page.' That's how an ad can track your identity and behaviors."

This synthesis, which the seeker could not have produced at minute zero, marks the session's first transition from Exploration to Collection in Kuhlthau's (2008) ISP terms. At minute 16:48, the seeker returned to the privacy.net article set aside at the session's opening and now read it deeply rather than skimming, extracting specific structural content including the cookie syncing mechanism and the limits of VPN plus fingerprinting. This is the focused, selective reading that Kuhlthau identifies as characteristic of post-Formulation behavior.

The second major phase, from approximately minute 23:42, shifted to network-layer questions. A Tor article produced a structured comparison of VPN and Tor threat models, clarifying a distinction the seeker had been treating as interchangeable: "Tor is more about anonymity, whereas a VPN is more concerned with privacy." A DNS article at minute 42:51 produced the second formulation wave, the DNS shock described in the introduction, which extended the conceptual map from the browser into the network layer. The session closed at exactly minute 60:00 on the timer, with an inconclusive query on ad network targeting, constrained by the time limit.

Analysis

Information Need and the Progression of Questioning

Taylor's (1968) four-level model provides the most direct theoretical purchase on how this event began. The seeker arrived at the conscious level, able to state the problem in rough terms but unable to

formalize it into a retrievable question. The opening query bears the marks of this stage: "what does a website learn about me when I visit" is broad, nearly definitional, and reflects an attempt to identify vocabulary before asking a targeted question.

The progression across the session is documented with unusual precision and is best illustrated by the queries themselves. At minute zero, the seeker could only formulate: "what does a website learn about me when I visit." By minute 46:22, after the DNS section, the query had become "does my ISP see my DNS queries even with HTTPS." The second query is impossible without knowing what an ISP is, what DNS is, how HTTPS encrypts, and why the gap between those two systems matters--all knowledge that did not exist at session's start. Showing these two queries in sequence enacts the Taylor progression.

The session also exhibits what Taylor describes as compromised formulation in specific moments. At minute 2:39, when the URL tracking demonstration failed to communicate its output, the seeker accepted a partial understanding rather than chasing URL parameter mechanics: "Nothing was supposed to change. That's a bit confusing. But I guess it's just to show how the URL itself can carry tracking info." At minute 38:02 and again at 54:45, the seeker encountered the term "cache" in sources explaining DNS but explicitly deferred it rather than pursuing it, choosing to preserve cognitive bandwidth for the primary line of inquiry. These are rational compressions, or compromises in Taylor's specific sense, between what the need demands and what the constraints permit.

This progression maps closely onto Dervin's (1992, as discussed in Savolainen, 2006) Sense-Making framework, which models information needs as "gaps" in the seeker's understanding that interrupt forward movement. The seeker's inability to evaluate tool recommendations without understanding the underlying mechanisms constituted precisely such a gap: not a gap in knowledge of a separable external subject, but a gap in the conceptual vocabulary required to make sense of information already being encountered. What this session illustrates, extending Savolainen's contribution beyond what the draft's original formulation captured, is that the gap is not static. It reshapes as the seeker moves through it. The gap at minute zero was formless. By minute 11:55 it had narrowed to a structural question about cookies and requests. By minute 42:51 it had shifted entirely into network-layer territory. Savolainen's (2006) rendering of Dervin's gap concept is especially apt here: information use is understood as gap-bridging, and the session succeeded to the extent that it built bridges, even partial ones, between the seeker's pre-existing understanding and the technical language of the domain.

The gap at session's end of "does my ISP keep logs, and under what circumstances does it share with law enforcement", was not a gap that existed at the start. The session created it. This dynamic reshaping is, in Savolainen's terms, the actual work of seeking.

Anomalous States of Knowledge and Iterative Formulation

Belkin's (1980) concept of the Anomalous State of Knowledge (ASK) illuminates a feature of this event that Taylor's model does not fully capture: the seeker often knew something was wrong with their understanding before they could say what it was. The early encounter with VPN comparison articles is illustrative. The seeker recognized that the article's terminology was meaningful and that understanding it would matter, but could not yet construct the query that would resolve the confusion. This is the condition Belkin describes as anomalous: the seeker has an incomplete, inconsistent, or incoherent state of knowledge, but cannot specify the information required to correct it because that specification would require the knowledge currently absent.

At minute 5:23, reading the Usercentrics article on pixels, the seeker noted that pixels are stored on a server the user cannot access, "which means I cannot see it, clear it, or even know what's there," and described the realization as "kinda scary." This is the affective signature of an ASK moment. The prior assumption, that what is on the user's device constitutes the relevant tracking surface, was being updated, and the update was felt before it was fully articulated. Belkin's claim that anomalous states are registered affectively before they are cognitively specified is on display in this passage.

The DNS moment at minute 42:51 is the second and most consequential ASK instance. The seeker had been carrying an implicit model: HTTPS protects browsing from external observation. Encountering the DNS resolution diagram forced that model into visibility by contradicting it. The realization that "every single time I type a domain name...it has to send a request to another server just to get the address of the server I want to visit...my ISP has a log of every domain I've ever typed" could not have been specified in advance, because specifying it would require already knowing what was wrong with the prior model.

ASK also explains the session's branching structure. When a VPN comparison article introduced DNS-over-HTTPS without explanation, the seeker did not flag the term and continue; the encounter triggered a felt incapacity that redirected search entirely. This is Belkin's prediction that anomalous states generate non-standard, exploratory queries rather than straightforward lookups. The seeker was not looking up what DNS-over-HTTPS means in the way one might look up a definition; they were attempting to understand what changed about their observable footprint when a setting was adjusted, which is a different kind of question requiring a different kind of source.

The Information Search Process: Stages, Affect, and Formulation

Kuhlthau's Information Search Process (ISP), as revisited by Kuhlthau, Heinström, and Todd (2008), offers the most comprehensive model for tracking the shape of this event as a whole. The ISP proposes six stages: Initiation, Selection, Exploration, Formulation, Collection, and Presentation. Each

stage carries characteristic cognitive, affective, and physical dimensions. The session's trajectory maps onto these stages with notable fidelity.

The event opened in the Initiation stage: the seeker had decided to act on an accumulated discomfort, and the affective register was one of mild apprehension mixed with resolve. The Selection stage was brief and informal; the decision to use DuckDuckGo rather than Google was itself a form of topic-consistent source selection, reflecting an awareness, however rough, that the choice of search engine was itself a variable in the privacy domain being studied. The Exploration stage dominated the first twenty minutes and was characterized by the confusion, frustration, and sense of incompetence that Kuhlthau describes as typical--sources used unfamiliar terms, explained things at the wrong level of abstraction, or presupposed technical background the seeker lacked. The affective quality of this phase was one of mild but persistent low-grade stress, occasionally spiking when a source seemed important but was too technical to follow.

Formulation is marked by a shift from diffuse searching to focused collecting, and that shift occurred at minute 11:55 with the cookie mechanism. Then again, at minute 42:51 with the DNS realization, marked the second formulation wave, extending the map into a new technical domain. The diagnostic evidence is behavioral: at minute 16:48, the seeker returned to the [privacy.net](#) article set aside early in the session and read it deeply and selectively; extracting, not exploring. That is collecting behavior, not exploring behavior, and the shift in the think-aloud record is audible as the commentary becomes evaluative rather than exploratory. Kuhlthau's prediction is that focused collection follows formulation, and the seeker's behavior at 16:48 confirms the earlier formulation.

The affective dimension is also well-documented in the transcript. At minute 1:20, reading [givemeyourdata.org](#): "I'm impressed." At minute 5:23, on pixel storage: "kinda scary." At minute 33:32, after extended reading on ISPs, advertisers, and hackers: "It's definitely a lot. I hope I'm not missing too much because I think I can feel myself getting a bit overwhelmed." This last entry is the clearest signal of what Kuhlthau and Heinström describe as affective volatility in the late Exploration phase, where cognitive load produces a felt risk of abandonment. The recovery that followed--explicitly identifying three take-away points and returning to a previously bookmarked source--is also Kuhlthau-predicted behavior: the seeker restructured the task rather than abandoning it.

Source Selection, Cognitive Authority, and the Adversarial Information Landscape

The source-selection dimension of this event is analytically distinctive because the topic itself creates unusual conditions for source evaluation. The landscape for privacy and tracking explanations includes rigorous technical writing, advocacy content with a point of view, outdated material describing mechanisms that no longer apply, and corporate explainers that are accurate about some things and

conveniently vague about others. Savolainen (2008) describes the concept of information source horizons: the ordered space of sources, from most to least preferred, that a seeker brings to a given need. This seeker's horizon was shaped by several factors: an orientation toward technically rigorous and open-source-aligned sources (favoring EFF, Privacy Guides, and similar communities over corporate explainers), a disposition toward skepticism about sources with an obvious commercial interest in the framing, and an awareness that the landscape included outdated material describing mechanisms that no longer apply.

Rieh's (2002) framework for judgment of information quality and cognitive authority in web environments maps directly onto the evaluation challenges encountered. Rieh distinguishes between predictive quality judgments (made before or during reading, based on surface features like site design, author credentials, and URL structure) and evaluative quality judgments (made after reading, based on content accuracy and internal consistency). In this session, the seeker relied heavily on predictive judgments: Privacy Guides and EFF were assigned high cognitive authority early, based on their community reputation in the OSINT and privacy spaces, their transparent editorial policies, and the absence of monetization signals. By contrast, several results from corporate security vendors were quickly dismissed, not because their content was necessarily wrong, but because their framing was predictably oriented toward selling a product. The seeker was aware of this as a bias and noted it in the think-aloud: "these people want me to buy a VPN, which means they have an incentive to overstate the threat."

The session also produced an observation that extends Rieh's framework. At minute 10:02, querying "how often are cookies sent to server," the seeker noticed that the result page was dominated by MDN, Stack Overflow, and developer documentation, with almost no compliance vendors: "This might be the first query this has happened for. I wonder why." The seeker's own answer that "how often are cookies sent" is a protocol question that surfaces developer audiences by default, identified that the composition of the result page as a whole, before any individual document is evaluated, is itself information about the kind of answer the query will produce. Predictive judgment, in other words, can operate at the level of the result page as a whole, not only at the level of individual sources.

Lu and Yuan (2011) identify a tension in source selection between quality and accessibility: people routinely prefer accessible sources over demonstrably higher-quality ones when the accessibility gap is large enough. This dynamic appeared in modified form here. The seeker had access to peer-reviewed technical literature (through library databases) but did not consult it during the session, defaulting instead to community-maintained resources that were immediately accessible via search. This was not a failure of judgment so much as a recognition that the seeker's goal was conceptual orientation, not comprehensive technical mastery, and that peer-reviewed sources on network protocols would have been more precise but also more demanding than the session's purpose required. The quality-accessibility

tradeoff was in other words made implicitly, appropriate to the session's conceptual information need rather than technical goal.

Exploratory Search and Berrypicking

Marchionini (2006) distinguishes between lookup search (finding a known item or fact) and exploratory search (learning about or investigating an unfamiliar domain). This session was exploratory throughout, and its structure follows Bates's (1989) berrypicking model with documentary precision. The actual berrypicking chain is traceable in the transcript. The opening query surfaced givemeyourdata.org, which mentioned UTM parameters, which sent the seeker to "UTM parameters"; that search surfaced marketer-facing documentation, which noted `gclid` and `fbclid`; querying those produced Google Support's advertiser help page. Returning to the original line, "tracking pixel vs cookie" surfaced Usercentrics; Usercentrics introduced tags; [learnwebanalytics](http://learnwebanalytics.com) added a diagram but left unanswered what "request" meant; "how often are cookies sent to server" surfaced MDN; MDN locked in the cookie mechanism; the seeker then returned to privacy.net, now equipped to read it deeply. This is six query reformulations driven by encountered material.

Such chaining behavior, identifiable in Ellis et al.'s (1993) characterization of information-seeking patterns, was the dominant structural feature of the session. The Tor article linked to a VPN comparison, which introduced DNS-over-HTTPS, which produced a sub-search, which surfaced the Techlore forum thread, which included a Cloudflare link on DNS encryption. The seeker navigated this chain without a pre-specified destination, which is precisely what berrypicking predicts.

Information Overload, Anxiety, and the Limits of the Landscape

Bawden and Robinson (2009) describe information overload not merely as the presence of too much information but as the experienced inability to process available information within the constraints of time, cognitive capacity, and purpose. The privacy and tracking domain creates conditions that are particularly favorable to this experience: the landscape is technically dense, politically contested, commercially incentivized in conflicting directions, and updated at a pace that quickly dates any synthesis. Early in the session, the seeker encountered this directly when over halfway through, a search returned dozens of results, many of them redundant, several contradicting each other, and most insufficiently specific about the threat models they addressed. The affective response was what the original draft's think-aloud record describes as an "urge to close the tab," a characteristic avoidance behavior consistent with Kuhlthau's Exploration stage and with the information anxiety Bawden and Robinson discuss.

Bawden and Robinson also distinguish between overload and information avoidance, and the session contains evidence of both. The deliberate deferral of "cache" at minutes 38:02 and 54:45 is selective avoidance--a rational choice to preserve cognitive bandwidth. The willingness at minute 5:23 to press into the pixel realization, described as "kinda scary," is the opposite--the seeker moving toward discomfort rather than away from it. These are different responses to different kinds of overload, and their coexistence in a single session suggests that overload management is selective and strategically calibrated, not a uniform retreat from the topic.

The seeker responded to this overload by narrowing scope explicitly: rather than reading across the breadth of the results, the seeker selected two sources and read them deeply, accepting incompleteness as the cost of cognitive manageability. This is a rational adaptation, and one consistent with Chatman's (1996) observation that people operating under uncertainty may prefer incomplete but manageable information over comprehensive but overwhelming information. The deliberate scope limitation stated before the session began, such as exclusion of action-oriented recommendations and tool comparisons, was a deliberate methodological choice that functioned as a pre-emptive overload mitigation strategy.

What Was Learnt, and What Remains Open

The session produced meaningful gains in two areas. First, the seeker developed a working model of the tracking ecosystem: first-party cookies set by visited sites, third-party cookies and tracking pixels placed by advertising networks, fingerprinting as a stateless alternative to cookies, and data brokers as the commercial layer that aggregates and sells profiles compiled from multiple sources. This structural map did not exist at the session's start. Second, the seeker developed a more differentiated understanding of privacy mechanisms: browser-level controls like uBlock Origin address third-party content loading; DNS-over-HTTPS changes what the DNS resolver can see but does not encrypt content; a VPN shifts the observation point from the local ISP to the VPN provider. Each of these insights represents gap-bridging in Dervin's sense.

The session also generated one design principle--that invisible processes must be explicitly named for demonstrations to work--what Marchionini calls the learning output of exploratory search. The output was not only answers but a transferable insight.

What remained open was the action dimension: the seeker had deliberately excluded tool selection from the session's scope but acknowledged in the final think-aloud commentary that the conceptual understanding gained would now make tool-selection sources, such as Privacy Guides' recommendations pages, legible in a way they had not been before. This is a characteristic outcome of successful exploratory search: not resolution, but the acquisition of enough conceptual structure to make the next phase of seeking tractable. Saracevic's (2007) multi-dimensional relevance framework helps

articulate what "success" means here. The most relevant material was not the most topically precise but the most cognitively and situationally relevant, sources that mapped onto the seeker's existing knowledge and addressed the purpose at hand, even where they fell short of technical completeness. The question map appended to the raw data--with clusters around "what actually happens when I visit a website" and "who can see what I am doing online"--could likely not have been produced at the session's start.

Reflections and Future Considerations

One dimension this analysis has not yet explicitly engaged is the connection between this personal seeking event and the professional information practice context within which it arose. The OSINT coursework named in the introduction as a motivating pressure is not incidental. The same source-evaluation skills that an OSINT investigator uses to assess the credibility of a tip--distinguishing technically accurate content from confident-sounding advocacy, assessing whether a source's incentive structure shapes its framing--are the skills the seeker applied to distinguish Privacy Guides from a vendor blog. The commercial data brokers catalogued by EPIC, encountered in the session, are the same brokers an OSINT investigator might query when building a profile of a subject. The privacy and tracking domain, examined from the subject's perspective, is the mirror image of the same infrastructure examined from the investigator's perspective. This symmetry makes the session more than a personal learning exercise, but also a concrete encounter with the information ecosystem that the course's investigative work also inhabits.

Several aspects of this event deserve attention for future similar needs. The absence of consultation with other people was notable. Sonnenwald's (1999) concept of information horizons includes people as a primary resource type, and the seeker had access, in principle, to peers with technical backgrounds. The decision not to consult them was partly practical (the session was time-bounded and self-directed) and partly affective: there was a sense that asking would require articulating the need in a way that exposed the extent of prior ignorance. This is consistent with findings in the literature on the social dimensions of information seeking, where face-protecting concerns shape source selection. In retrospect, a thirty-minute conversation with a technically knowledgeable peer might have produced faster convergence than fifty minutes of web searching.

The choice of DuckDuckGo over Google was not merely thematically apt but introduced an uncontrolled variable. Whether DuckDuckGo produced meaningfully different results on privacy-related queries than Google would have is an empirical question the seeker noted but did not resolve, though it is worth flagging as a variable that future observers of similar events might control for more explicitly.

Finally, the think-aloud method, while effective for capturing real-time cognition, introduced a small interference. Narrating reasoning sometimes delayed the reading of a source long enough for the cognitive context that made that source relevant to partially dissipate. Future applications of this method in self-observation contexts might benefit from shorter verbal check-ins rather than continuous commentary.

Conclusion

This event was a recognizable instance of exploratory information seeking, oriented toward conceptual rather than instrumental goals, and conducted in a domain whose structural features, including competing incentives among sources, rapid obsolescence of technical details, and the recursive quality of seeking information about one's own informational exposure, created conditions favorable to overload, detour, and evaluative difficulty. Applied across the event, Kuhlthau's ISP, Belkin's ASK, Taylor's need-level progression, Dervin's gap-bridging framework, and the source-evaluation literature each illuminated a different dimension of what took place. The session's intellectual arc followed Taylor's need-level progression from naive to formalized questioning, moved through the affective stages of Kuhlthau's ISP, was structured by the iterative branching Bates identifies as berrypicking, and was shaped at specific junctures by the anomalous states Belkin describes. Source evaluation was adversarial, requiring Rieh's predictive and evaluative judgment frameworks simultaneously. The outcome arrived not at resolution but at a stable foothold: a conceptual map detailed enough to make the next phase of seeking, and the action it might eventually support, genuinely possible.

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Appendix

Raw Data

Pre-Session Notes

Topic: tracking and privacy protection. *Goal:* build a working understanding of how data is collected as I move around the internet and how the most common exploitative or protective mechanisms actually work.

Setup: Timer set for 60:00. Voice and screen recording capture on; I will be employing a think-aloud approach, in which I will commentate whatever I am thinking in the moment as I conduct my searches and later transcribe with timestamps and thoughts--so as to not spend too much time on the note-taking during the actual search session. I have a notepad beside me to jot down any concepts or terms I want to come back to. I will polish and annotate the transcript with additional notes afterwards if deemed necessary. I like to switch between using DuckDuckGo and Google as my primary search engine. Due to the nature of my searches, it seems fitting to proceed my search on DuckDuckGo, using Mozilla Firefox as my standard browser with my normal extensions (uBlock Origin, Adblock, Multi-Account Containers, Privacy Badger, Zotero).

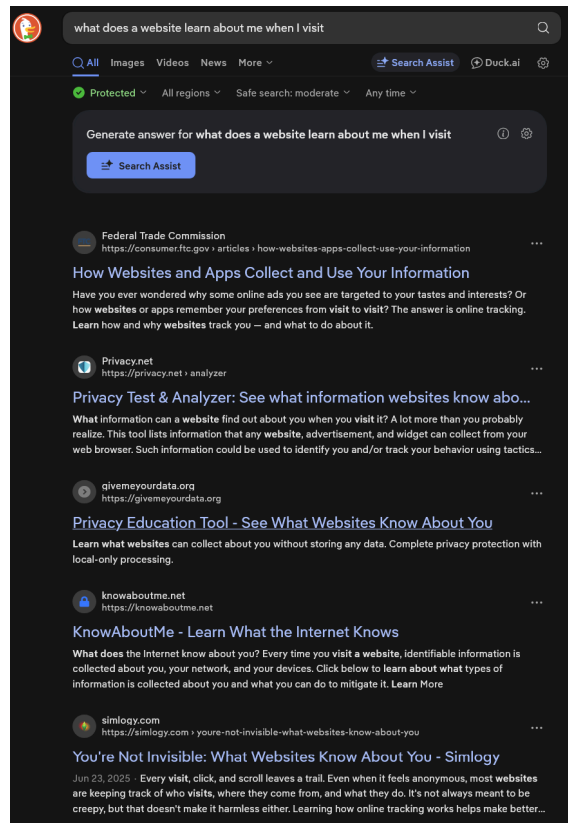
I have drafted a few general top-level questions to start with so as to not waste time from the get-go thinking about what to write. I hope to develop narrower sub-questions in scope, as I understand what is possible to even ask. Examples:

1. What actually happens when I visit a website?
2. Who can see what I am doing online, and when?
3. What is tracking, mechanically?
4. What do privacy tools actually do?
5. How do entities in the tracking ecosystem relate to one another?

A note on what I already know going in: I am somewhat familiar with vocabulary like DNS, cookie, fingerprint, HTTPS, VPN, ISP, metadata, tracker, proxy. I could tell you, roughly, that cookies are "for tracking" and that a VPN "hides your location." But that recognition is surface-level. I know what role each word plays in a sentence. I do not know what the thing actually does, how it does it, or why one of these things is meaningfully different from the next. The session is meant to convert recognition into mechanism for as many of the central concepts as I can fit into an hour.

0:00 *[Opens new tab, navigates to DuckDuckGo]* Okay. Starting. The thing I actually want to understand is... what happens when I visit a website. Like, I know cookies exist. But I cannot actually describe the sequence of events that produces the thing where I see an ad for something I mentioned in conversation. Let me start there. First query.

0:18 [Types 'what does a website learn about me when I visit' into DuckDuckGo] Alright I'm getting some different options I can follow. [Opens the FTC site in a new tab] This one is a pretty official source. [Skims through it.] Not too long either. But it's looking pretty high-level; clearly more occupied with actionable recommendations rather than actually detailed explanations.



0:37 [Exits out of FTC and clicks on privacy.net] Oh this is just a test. [Clicks on Home page] Ah but there are some actual informational blog posts that might be helpful. [Opens two articles in new tabs. One is on cookies and the other is on the Tor Network.] Gonna come back to these later I think. The blogs might be easier to read but I'm a bit wary about the accuracy and neutrality of the information.

0:52 [Opens givemeyourdata.org] This site looks scary but worth a shot... Wow they just straight show me all the details available on my device. This is so cool. And it has really helpful explanations at the bottom too. [Navigates to home page] This actually looks really similar to what I hope to eventually make. I really want to make my own information security guide once I have a better understanding of it myself. Let me go through this more in-depth to figure out what I want to search next.

1:20 [Reading] I'm impressed. I really appreciate that they very directly list the answers to my original question... Dang I didn't know that it's standard for a site to automatically receive 'referrer' information about how I got to that site. I thought that would have been more encrypted with DuckDuckGo, but I guess that's not one of the capabilities of the engine since it can obviously still be detected. Interesting. [Tries the URL Tracking Demonstration] It said I could

add some parameters to my current URL to see how websites track your traffic source, but I'm not really sure where I'm supposed to be looking or if there are even any differences from before. *[Annoyance note: the demo invited me to do something and didn't tell me what success looks like. The page apparently expects me to know what a UTM parameter is and what it would change. I do not.]*

2:03 *[Reading through "How Real Websites Track You", "How Consent Levels Work", and "Real-World Privacy Risks" sections]* The site is honest that what it's showing me is a subset of what commercial sites actually do. They list some stuff that I think would be helpful next steps to explore in my search.

2:18 *[Tabs back to DuckDuckGo. Types 'UTM parameters']* Ok I figured this is what it meant, but I still don't get what I was supposed to see changed, if anything... Interestingly enough, much of the search results I'm receiving seem to be explaining UTM parameters from the perspective of someone setting them up to track their own marketing campaigns. The explanations exist, they're easy to find, but they're written for the people adding the tracking, not the people being tracked...

2:39 Ok so what I'm getting from this is that nothing was supposed to change. That's a bit confusing. But I guess it's just to show how the URL itself can carry tracking info. *[Annotation for my own future guide: when demonstrating something invisible, name the invisibility. Don't let the user think the demo doesn't work.]*

2:48 *[Types 'gclid fbclid what are they']* The UTM demonstration earlier also had these in their parameters so I'm just curious... Huh. I mean it's a simple explanation, but what's more interesting is that the titles are basically all variations of "What is GCLID and how to use it for tracking conversions / measuring ROI / proving marketing success." The information for setting up tracking is everywhere, but the information for understanding what's happening to you seems to be much harder to surface. *[Note: Ten results explaining what the thing is, all of them written for the people deploying it. Even Google Support is a marketer-facing help page. The privacy-side explanation of these identifiers does exist somewhere, but DuckDuckGo's ranking, presumably reflecting the broader web's, has none of it on page one.]*

3:20 *[Opens Google Support (<https://support.google.com/google-ads/answer/9744275?hl=en>) result.]* ...So when a Google Ad gets clicked, Google automatically appends an id value to the destination URL. The destination site logs the click with that ID. Later, when that ID shows up associated with a purchase or signup, Google can match the original ad click to the eventual outcome... It's not explicitly said here, but since Google also runs services I'm signed into, like Gmail and YouTube, Google can associate that click with my actual logged-in identity and probably make a comprehensive profile of me.

4:31 Alright. Going to back up and finish what the demo originally raised. *[Tabs back to DuckDuckGo. Types tracking pixel vs cookie]* I came down the URL-parameter rabbit hole

because the demo confused me, but what I actually wanted to explore based on what I learned from that site was tracking pixels and third-party cookies.

4:50 Hm. These are mostly consent-and-compliance vendors, so these are mostly written for site operators. They come more so from an angle of legal compliance, not visitor protection. But the explanations of mechanisms, because they have to be accurate enough to satisfy regulators, are usually pretty good. *[Opens a few pages. Starts with the Usercentrics (<https://usercentrics.com/knowledge-hub/pixel-cookie/>) result]*

5:23 ...Good consolidated explainer, even though it's pitched at site operators trying to stay compliant with privacy regulations, rather than at me as a visitor.

Three things in this article are catching my attention more than the rest.

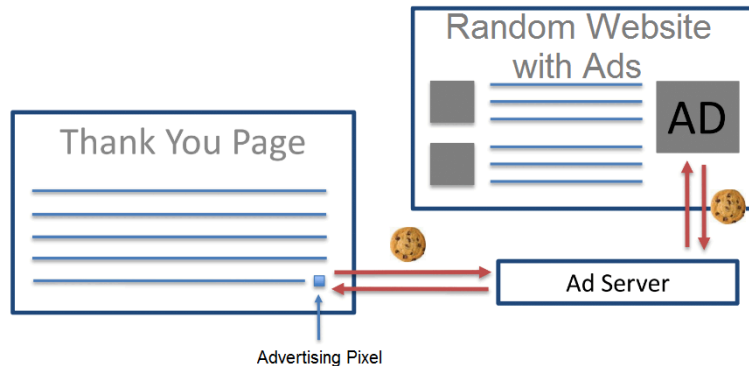
First, there are different types of pixels. Engagement pixels (time on page, clicks), retargeting pixels (track visitors so they can be shown ads on other sites), and conversion pixels (record specific actions like a signup or purchase). Pixels are events fired to a server when an embedded resource (often a transparent 1x1 image) loads, enabling cross-site tracking. I had been thinking of "the tracking pixel" as one thing. Well that answers one of the questions I had I think, "why am I seeing ads for this site I visited once." There are also many kinds of cookies, though this I was more familiar with already. Session cookies last only while you're actively browsing, while persistent cookies continue to function for a specified amount of time and are stored on your device. Meanwhile, first-party cookies come from the website you're visiting, and third-party cookies come from external sources like ad networks.

Second, within the comparison table, it says cookies are stored "in the user's browser" and pixels are stored "on an external web server." With cookies, the data is on my device, which means in principle I can see it, clear it, or block it. With pixels, the data is on someone else's server, which means I cannot see it, clear it, or even know what's there. I suppose it makes sense that pixels are invisible because they're stored somewhere I have no access to. But that's also kinda scary. I know clicking strange links isn't a good idea because you don't want the page to load in and let the pixels start firing data about you, but I didn't fully realize that even opening an email still exposes you the same way.

Third, the article flatly says pixels are "harder for users to block" because "doing so often requires them to use ad blockers or disable automatic image loading." That's a clean explanation of why my uBlock Origin matters in a way I didn't fully understand before. It's not just blocking annoying ads; it's blocking the image-load that is the tracking event. The block prevents the request from being made, which means the server never gets to log it. I'm not sure if this still applies even when I'm browsing through my email though.

9:14 *[Switches to learnwebanalytics (<https://learnwebanalytics.com/whats-the-difference-between-a-cookie-a-pixel-and-a-tag/>)]* This one also includes information about tags... Apparently tags and pixels are often used

interchangeably, but a “pixel is always going to be a tag, but a tag isn’t always going to be a pixel.” All a pixel tag is is an element on the page and all of the attributes relating to that pixel. I wonder what implications that might have for the user though... I also really like this diagram they added. I do tend to be more of a visual learner sometimes.



The article says cookies "are sent back to the server with each request." But I'm a bit unclear about what exactly it means by "request."

10:02 [Tabs back to DuckDuckGo. Types 'how often are cookies sent to server'] ... Cleaner result page than the last few. Search Assist gives me the answer up top: cookies are sent with every HTTP request to that server, as long as the cookies are relevant to the domain and path of the request. Sources cited include Wikipedia and Stanford. There's also a sidebar pulling a Stack Overflow answer that says the same thing more bluntly: "as long as the URL requested is within the same domain and path defined in the cookie... the cookie will be sent for every request." I'm not too confident on what exactly this means still, but I'm more interested in the nature of the search results I received. MDN at the top, Stack Overflow second, then DEV Community, then mostly developer-documentation sites (http.dev, GeeksForGeeks, Wikipedia, MDN again via devdoc.net). Almost no compliance vendors or marketing sites. This might be the first query this has happened for. I wonder why. [Note: "how often are cookies sent" is a how-does-the-protocol-work question, which might be the reason for having a developer audience by default.]

11:55 [Opens MDN Using HTTP cookies.] ... Ok let me try to break this down. When my browser is about to make any request, it checks: do I have any cookies whose domain matches this request's domain and whose path matches the URL? If yes, it includes them in the Cookie header automatically. This happens on every request, not just page loads. So when a single page loads and pulls in 50 different resources from 30 different domains, my browser is making 50 requests, each one carrying whatever cookies match that particular domain.

The diagram in my notes makes more sense now. When I land on a "Thank You" page that has an advertising pixel on it, my browser fetches the pixel from the ad server. That fetch is a request. Because I've previously visited some other site that uses the same ad server (via third-party cookies), my browser already has a cookie for the ad server's domain. So the pixel-fetch request automatically carries that cookie. The ad server now knows: "the same

browser that I previously identified by this cookie is now on this Thank You page." That's how an ad can track your identity and behaviors.

Something new from this page is that cookies have flags that change their behavior, like Secure, HttpOnly, and SameSite. I appreciate how this site breaks down what the code actually looks like and provides examples for what implications different attributes actually have. I'll have to save this one.

16:48 *[Navigates back to previously opened [privacy.net](https://privacy.net/stop-cookies-tracking/) blog post "How cookies track you around the web and how to stop them" (<https://privacy.net/stop-cookies-tracking/>)]* I think it's time to stop keeping these tabs on the side and actually read through them. ...Wait this is actually super comprehensive. It even tells you the answer to a question that was left unanswered on our previous site--sometimes cookie can be accessed through JavaScript, which is why disabling it can prevent "cross-site scripting (XSS) attacks that can be used to steal login credentials and modify cookies for nefarious purposes." It also expands on what exactly can be seen through pixels (otherwise known as web beacons, though this is the first time I've seen the term used this session). For example, when a user opens an email with an embedded pixel tag, "the email must load a resource from a third-party. When this happens, the resource can request the recipient's IP address, timestamp, type of browser, and whether the resource owner already set a cookie in this browser. Like cookies, the server can store all of this information and associate it with the user's unique tracking ID." It also highlights that "Even if you connect to a VPN to hide your IP address and block tracking cookies, all of the other information can form a combination so specific that the resulting profile can only plausibly belong to a single person or small group of people. Attempting to alter your browser settings and install more plugins only makes you stand out more." I don't know why I was hesitant to read from privacy.net. It's clearly made for users searching for information like me, and I really appreciate that it also explains how different stakeholders like ad companies can use practices like "cookie syncing", where two advertising companies partner up or acquire one another to combine advertising data sets, creating more accurate and comprehensive tracking profiles. This is what helps them compete with Google, but has serious implications on user privacy. I think this is a good transition point into questions on the network level.

23:42 *[Navigates to previously opened [privacy.net](https://privacy.net/what-is-tor/) blog post "Everything you wanted to know about Tor but were afraid to ask" (<https://privacy.net/what-is-tor/>)]* I think I already know enough about IP addresses that asking about them directly would just regurgitate the same foundational information. This Tor article looks really interesting though.

24:30 Okay, this is denser than I expected and I want to slow down and pick out what's actually new to me versus what is vocabulary I was already half-carrying.

Tor stands for The Onion Router. It was originally developed by the US Navy in the mid-1990s. Interesting that an anti-surveillance tool was built by the military.

The mechanism: outgoing traffic gets bounced through at least three randomly-selected nodes (entry, middle, exit) before reaching the destination, and the traffic is encrypted in layers (like an onion!), one for each hop. Each node peels off one layer. The destination only ever sees the exit node's IP, not mine.

The trust model is super interesting. The article calls Tor "trustless" because no single node knows both who I am and where I'm going. The entry node sees my real IP but not my destination. The exit node sees my destination but not my real IP. The middle node sees neither end. Compromising one node tells the attacker very little. Compromising all three at once is what would break the anonymity, and the random selection makes that hard.

In comparison, with a VPN, one party (the VPN provider) sees both ends. They can still log who I am and where I went. The article states "Tor is more about anonymity, whereas a VPN is more concerned with privacy." Privacy is hiding contents from outside observers. Anonymity is severing the link between identity and activity. Different goals, different mechanisms. I had been using these words interchangeably.

28:50 The downsides section is honest and explanatory, which I really appreciate. Tor is slow because traffic takes a long detour. ISPs can detect you're using Tor even if they can't see what you're doing, which can itself draw attention. The exit node operator can see unencrypted traffic passing through them, so HTTPS still matters even on Tor. And the article notes the "Silk Road takedown of 2013" as evidence that Tor isn't bulletproof against motivated state-level adversaries.

30:09 I'm really trying to comprehend this dynamic between which to use first, Tor or a VPN, and what is the difference. Tor-over-VPN: connect to VPN first, then Tor. ISP sees only the VPN connection. VPN sees only that I'm connecting to Tor. Tor entry node sees the VPN's IP, not mine. That's the recommended configuration, and the reasoning makes sense once it's laid out.

31:55 *[Clicks on another blog post "How to stop ISPs, advertisers, and hackers gathering information about your child" linked within [privacy.net](https://privacy.net/stop-isps-tracking-your-child/) (<https://privacy.net/stop-isps-tracking-your-child/>)]* I think the advice in here can be very relevant to what I want to know...

33:32 Ok I think there are three points I want to keep:

First, the article notes that ISPs in the US "have free reign to use browsing data to earn profits through advertising," and that some ISPs (Verizon and AT&T are named) document opt-out procedures. I had been assuming opting out of ISP tracking was either impossible or buried so deep that nobody does it. The article suggests it exists and is documented. Whether it's effective is another question, but the option is there. Worth checking later.

Second, the article distinguishes three different threat actors in its title: ISPs, advertisers, and hackers. They're often lumped together as "people who want your data," but the mechanisms by

which each one gets your data are different, and the defenses are different. ISPs see your traffic because it goes through them. Advertisers see your traffic because they're embedded on the pages you visit. Hackers see your traffic when they actively intercept it (man-in-the-middle on open Wi-Fi, phishing, malware). Three different threat models, three different sets of countermeasures. It's definitely a lot. I hope I'm not missing too much because I think I can feel myself getting a bit overwhelmed.

Third, using Facebook or Google to sign in to other apps gives those platforms visibility into where you're using their identity, and gives the third-party app access to your social profile data.

35:46 ...Something I want to note is that the action items are a mix of legitimate (HTTPS Everywhere, ad blockers, anonymous search engines, opt out of ISP tracking) and uncritically affiliate-driven (the VPN section reads like a sponsor placement, the mSpy recommendation is presented without naming that it's a surveillance product the parent installs on the child). The piece is from 2018 and clearly part of a content-marketing pipeline that ends in VPN affiliate links. Worth using for the threat-actor framing, but not really for tool recommendations.

36:31 I've been on this page for a while now. Let's move on. *[Tabs back to DuckDuckGo. Searches 'what is a DNS, how does it work, why does it matter'. Before spending much time on the page, retypes search query into 'what is a DNS, how does it work, why does it matter for privacy and security']* Oh there are a few sites I recognize like geeksforgeeks. *[Clicks (<https://www.geeksforgeeks.org/computer-networks/domain-name-system-dns-in-application-layer/>)]* Ok immediately upon scanning, it looks like there's just a whole bunch of vocabulary, like types of DNS Servers and Queries. I do appreciate how they seem to try to hit every component of a Domain Name System (DNS) and how it works, but I certainly don't think I'll be able to understand all of this at once.

38:02 *[Tabs back and clicks <https://www.privateinternetaccess.com/blog/what-is-dns/>]* This article is structured more like an FAQ, and they actually directly answer step-by-step "what's going on when you visit a website". It assumes understanding of "cache", but that could be another topic to do more in-depth research on. It also speaks on common types of DNS attacks and how they work, which I've decided is a format I also want to implement in my guide.

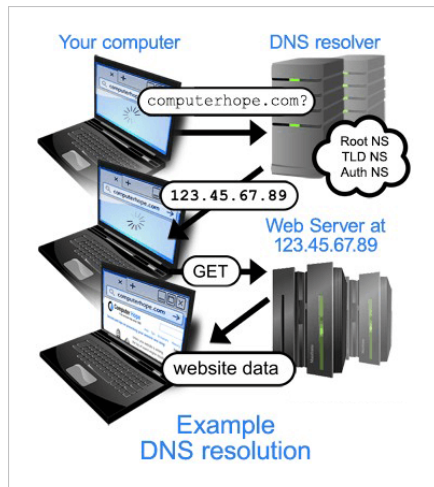
40:32 *[Tabs back and clicks <https://www.hostinger.com/uk/tutorials/what-is-dns/>]* Oh I'm already really liking this site. It has lots of diagrams and screenshots to show you examples of what everything typically looks like. I don't think I've seen many screenshots used in the other pages I've visited today. At this point I'm moreso skimming through the content for what it has, and not spending too much time trying to accurately comprehend everything being said...

42:51 *[Pauses skimming, slows down on a diagram of the DNS resolution chain]* Wait, actually I want to pause here. I keep seeing DNS described as "translates domain names to IP addresses" but I'm only just now registering what that actually means for me as a user... Ok so, every single time I type a domain name, before my browser can do anything else, it has to send a request to another server just to get the address of the server I want to visit. That request--the

DNS lookup--goes somewhere. By default, it goes to my ISP. Which means my ISP has a log of every domain I've ever typed into a browser, regardless of whether the page itself is encrypted. HTTPS protects the content after the connection is made. It doesn't protect the lookup that made the connection possible.

That's actually a really important gap in what I thought I understood. I was under the impression that HTTPS mostly solved the "people can see what I'm doing" problem. But if every domain lookup is visible to my ISP in plaintext, HTTPS is much less helpful than I thought at the network level.

45:08 [Types 'DNS resolver meaning' into DuckDuckGo and clicks on Computer Hope link (<https://www.computerhope.com/jargon/d/dns-resolver.htm>)]. This actually helps a lot because people kept talking about DNS resolvers, but I couldn't really figure out what they meant by resolver. I get now that it's just the actual Internet server that converts domain names into IP addresses, and that communication process is what is called the Domain Name System (DNS) resolution. I'm also loving this diagram.



ComputerHope.com

It even goes in more depth about what steps occur in a commonly-used Domain Name System, which I find very valuable in visualizing how all these terminology and concepts actually interact with one another.

46:22 [Types 'does my ISP see my DNS queries even with HTTPS' into DuckDuckGo] Yeah, I was just confirming. Several results say yes, unless you use an encrypted DNS specifically, like DNC over HTTPS (DoH). Reading from the AI Search Assist citing Vercara and a post made on a "Techlore" forum, I'm told that however, even with DoH, the organization controlling the DoH server can still see the domains you are requesting. [Opens both Search Assist citations] This guide to DoH (<https://vercara.digicert.com/resources/encrypted-dns-queries>) is pretty good, going over the limitations of DNS, history and basic mechanics of DoH, challenges of implementation, and what future adoption is looking like. It does ultimately conclude with Vercara as a "digicert" company advertising their 'UltraDNS' product to better "balance user privacy and network security." They explain that encrypted DNS traffic makes it harder to detect malicious domains, enforce acceptable use policies, or monitor for command-and-control (C2)

traffic. This is fair and probably important to note, but I think not going to be a huge consideration or priority for the kind of audience that tends to search for a guide like this.

47:46 [Switches to Techlore Forum 'What Your ISP Can See'

(<https://discuss.techlore.tech/t/what-your-isp-can-see/12157>)] I appreciate the step-by-step breakdown including examples of what they see in a literal manner. I'm going to copy some of the notes in here for reference.

1. When connected to HTTP (i.e No lock icon next to the URL)

- Your ISP can see everything as HTTP is unencrypted. So the website itself, images and/or any content you are connecting to is visible to your ISP.
- URL: The entire URL is visible; They can see the pages you are connecting to: example.com/page1, example.com/page2

2. When Connected to HTTPS (i.e Lock icon visible next to the URL)

- Your ISP can only see the website you are visiting, but nothing else.
- URL: Only the "top" level domain is visible to them; They cannot see that you are viewing page1 and page2 but they can see that you are connected to example.com.

3. Using a Different DNS Server

- The content visibility still depends on whether you are using HTTP or HTTPS; but now since you are using a different DNS Server, your ISP only sees the IP Address of the website you are connecting to.
- Note: They can still do something called a reverse DNS lookup from which they can see what website you are connected to.
Your ISP sees: 96.7.128.187 but not example.com (unless they perform a reverse DNS lookup).

4. Using a VPN

- Your ISP only sees that you are connected to the VPN's server but they cannot see what content you view or what websites you visit. Instead all your traffic is visible to your VPN provider.
- Everything listed in #1, #2, #3 applies to the VPN provider.

Additional Notes

- HTTPS = Military Grade Encryption
 - VPN ads love this term but all they are talking about is HTTPS. You are connected to this form using "Military Grade Encryption" regardless of you using a VPN or not.
- To avoid confusion: DNS is a separate system from HTTP and HTTPS. While DNS helps translate domain names to IP addresses, it doesn't control the encryption of the web traffic itself.

48:02 There's another comment from the original poster made in response to a question, linking to a Cloudflare site explaining DNS Encryption

(<https://blog.cloudflare.com/dns-encryption-explained/>). They put it simply, clarifying that

- Unencrypted DNS requests can be intercepted, modified or redirected by an ISP.
- Encrypted DNS cannot at any stage be tampered with.

48:40 [Tabs back to DuckDuckGo search and clicks on ISP Spy Check link

(<https://ismyispspying.com/blog/what-isp-can-see/>)] Just wanted to check this out. It looks pretty comprehensive. I especially like this table... Alright, in most of the sites about the DNS, they typically assume understanding of "TLS" and "SNI", which I've seen multiple times before, but have forgotten a lot of what I may have known.

The Full Visibility Table				
What you do	ISP DNS	ISP network	With DoH	With VPN
Visit example.com (HTTPS)	✓ sees domain	✓ sees IP + hostname (SNI)	X hidden	X hidden
Visit example.com (HTTP)	✓ sees domain	✓ sees full URL + content	X hidden	X hidden
Pages you visit on HTTPS	X	X content encrypted	X	X
Passwords you type	X	X encrypted	X	X
Every domain you look up	✓ full log	—	X encrypted	X hidden
Volume of traffic	—	✓ always visible	—	✓ sees only VPN server
Time of activity	—	✓ always visible	—	✓ only VPN server
Your real IP address	—	✓ always visible	—	X hidden

49:05 [Tabs back and types 'sni and tls'] Mmm. [Opens geeksforgeeks link

(<https://www.geeksforgeeks.org/computer-networks/what-is-server-name-indication-sni/>) and

cloudflare links (<https://www.cloudflare.com/learning/ssl/transport-layer-security-tls/> and

<https://www.cloudflare.com/learning/ssl/what-is-sni/>)] ...So server name is just the computer's

name. Unless the server is hosting a single domain with the same name as itself, "this name is not displayed to end users for web server." TLS (Transport Layer Security) is just a protocol that ensures the right connections are being made via certificates. A connection is initiated using a sequence called the TLS handshake, occurring between the user's device and the web server.

During this handshake, both servers

1. Specify which version of TLS they will use
2. Decide on which cipher suites (see below) they will use
3. Authenticate the identity of the server using the server's TLS certificate
4. Generate session keys for encrypting messages between them after the handshake is complete

[Separately copied some notes]

The TLS handshake establishes a cipher suite for each communication session. The cipher suite is a set of algorithms that specifies details such as which shared [encryption keys](#), or [session keys](#), will be used for that particular session. TLS is able to set the matching session keys over an unencrypted channel thanks to a technology known as [public key cryptography](#).

The handshake also handles authentication, which usually consists of the server proving its identity to the client. This is done using public keys. Public keys are encryption keys that use one-way encryption, meaning that anyone with the public key can unscramble the data encrypted with the server's private key to ensure its authenticity, but only the original sender can encrypt data with the private key. The server's public key is part of its TLS certificate.

Once data is encrypted and authenticated, it is then signed with a message authentication code (MAC). The recipient can then verify the MAC to ensure the integrity of the data. This is kind of like the tamper-proof foil found on a bottle of aspirin; the consumer knows no one has tampered with their medicine because the foil is intact when they purchase it.

SNI (Server Name Indication) is an extension included in the TLS handshake process in order to ensure that client devices are able to see the correct TLS certificate for the website they are trying to reach.

[More notes copied to not spend time explaining it out loud, but to keep for later reference]

The extension makes it possible to specify the hostname, or [domain name](#), of the website during the TLS handshake, instead of when the [HTTP](#) connection opens after the handshake.

More simply put, SNI makes it possible for a user device to open a secure connection with <https://www.example.com> even if that website is hosted in the same place (same IP address) as <https://www.something.com>, <https://www.another-website.com>, and <https://www.example.io>.

SNI prevents what's known as a "common name mismatch error": when a [client \(user\) device](#) reaches the right [IP address](#) for a website, but the name on the SSL certificate doesn't match the name of the website. Often this kind of error results in a "[Your connection is not private](#)" error message in the user's browser.

54:45 I wanted to come back to the cache thing eventually, but since I don't have much time left, I'll instead try to circle back to the original mechanism I was curious about. *[Tabs back to DuckDuckGo. Types 'how do ad networks decide what ad to show me']* Almost all of these are platform-side or advertiser-side. The explanations on the official sites for Facebook or Google AdSense, for instance, are very short, just showing the different types of targeting approaches that can be taken (ie. Contextual, Placement, Personalized, Run of Network) or their machine learning strategy.

<https://affvertising.com/ad-networks/how-ad-networks-work/> - this is more about how ad networks take an inventory from a wide range of publishers to allow for a centralized system connecting advertisers with publishers that have available ad space, tailored to be optimized towards users who are most likely to engage.

<https://internetunderstood.com/how-does-facebook-know-what-ads-to-show-you/> - blog written by a professional Facebook advertiser. just gives “assurances”, not very helpful.

58:10 *[Opens ITIF*

(https://itif.org/publications/2021/11/02/itif-technology-explainer-how-do-online-ads-work/)] This is probably a better source, being the only one not directly addressing the advertisers or publishers. It instead speaks more directly to the tracking technology. ...It does explain that contextual ads aren't just a simple substitute for personalized ads, since advertisers marketing niche products and services may struggle to find appropriate sites to use for contextual ads and many publishers will earn less revenue from contextual ads than targeted ads...

...These strategies for improving consumer privacy and preventing ad fraud are pretty cool. I think this one stands out most. “Google has introduced Federated Learning of Cohorts (FLoC), a new technology that allows advertisers to target users without receiving any of their personal data. Google's FLoC API uses algorithms to sort users into various cohorts based on interests determined by browsing history. These algorithms are computed locally on each user's device, meaning that there is no central server collecting behavioral data. Members of cohorts are kept anonymous, with cohorts represented by an ID number.”

60:00 Well I guess that's time. *[Sits back for a moment]* Not too bad. I want to finish off with some post-session reflection before I forget. *[Looks back through notepad]*

Ok well I came into this session with a pretty clear goal for what I wanted out of this, even if I wasn't sure exactly where to search. Since my aim is to eventually be able to create a technology and information security guide archive of my own (wikipedia/khan academy-style), I tried to also note down which sources I used and how well they delivered the information I was looking for. Was it accessible or hidden? verifiable or uncited? I wanted to be able to answer questions like how do we know this? what makes a website or tool really good or lacking in its ability to answer the questions one is conducting research with? In my guide, I want to advise on topics like how to navigate the internet without being tracked (where/how are all the sources one can be tracked from and why), comprehensively explaining the details behind various layers of security. Closing this search session off, I still have many questions left unanswered. I have some unfamiliar terminology underlined in my notes that I never ended up getting time to properly go back and understand.

However, there were some key points that I got a good understanding of that I think will help make the rest of the research a lot easier to comprehend.

- The full chain of actors from device to server and what each can observe
- The specific limits of HTTPS, DoH, cookies, pixels, and fingerprinting
- The difference between content blocking and tracker prevention

I also wonder how different my search results may have been if I used a different engine like Google instead.

Additionally, I think for next time it would be helpful to create a more dynamic question map. I had started out with a few general questions that I planned to narrow down through my search, but I think breaking it down visually like how I have below would make a lot more sense for

me--though this could only be done after having an understanding of what different terminology even means.

Example Question Map:

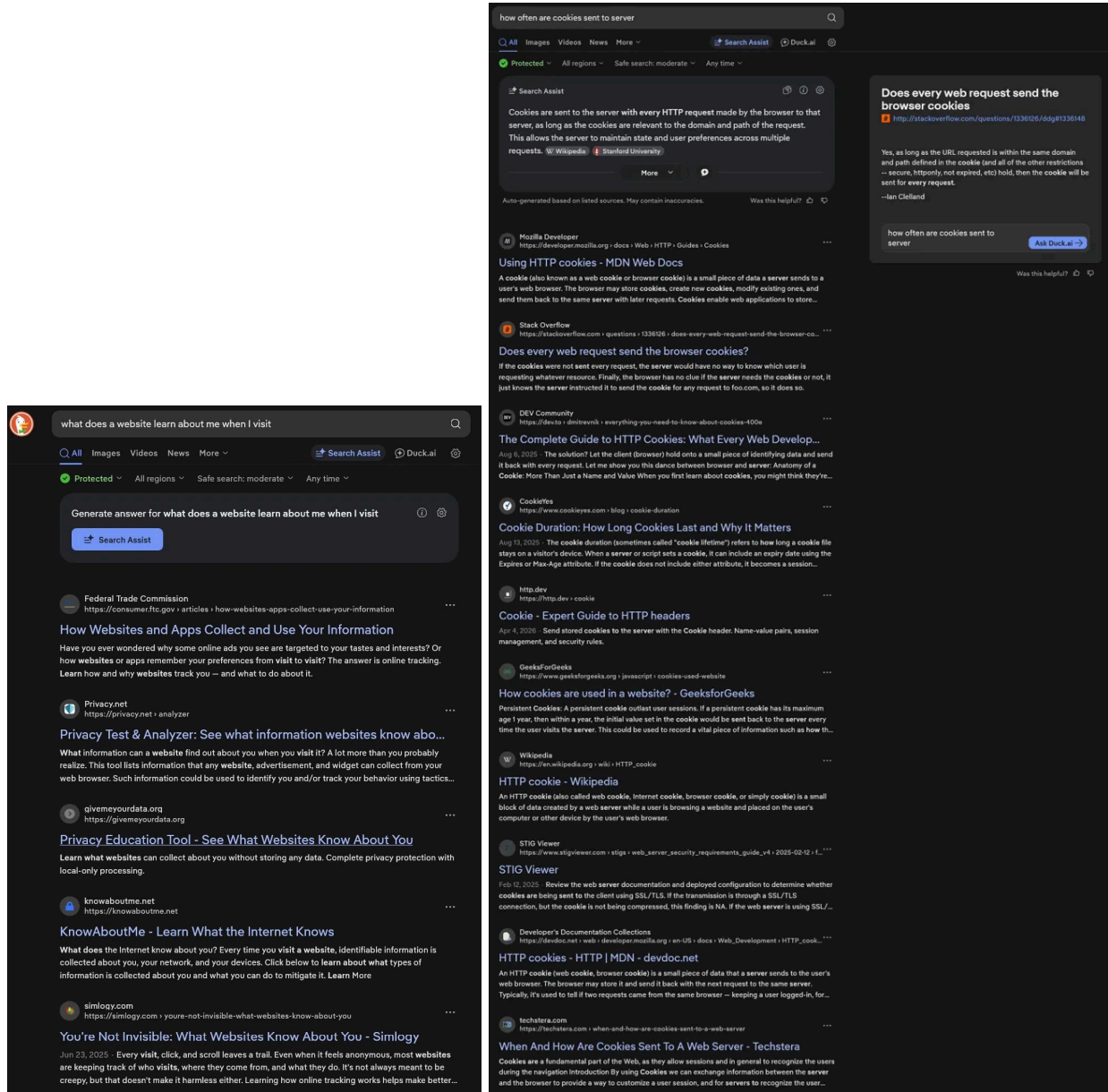
Cluster A - What actually happens when I visit a website?

- A1. When I type a URL and press Enter, what literally happens?
 - A1a. What is a DNS request, and where does it go?
 - A1b. What is an IP address, and what does it reveal about me?
 - A1c. What is a TCP connection, and who is involved?
 - A1d. What is HTTPS, and what does the "S" actually do?
- A2. What does HTTPS actually encrypt? What does it leave visible?
 - A2a. Can my ISP see the websites I visit if I use HTTPS?
 - A2b. Can my ISP see what I do on those websites?
 - A2c. What is Server Name Indication (SNI), and why does it matter?
 - A2d. What is TLS, and how is it different from SSL?
- A3. What is metadata, and how is it different from content?
 - A3a. What metadata is exposed even over an encrypted connection?
 - A3b. What can timing and volume of traffic reveal, even without content?

Cluster B - Who can see what I am doing online, and when?

- What can my ISP see?
 - B1a. Does HTTPS limit what my ISP sees?
 - B1b. What is Deep Packet Inspection, and when is it used?
 - B1c. Does my ISP keep logs? For how long?
 - B1d. Under what circumstances does my ISP share data with government or law enforcement?
- B2. What can the website I visit see?
 - B2a. What is my IP address telling the server?
 - B2b. What does the HTTP request header contain, and who reads it?
 - B2c. What can the website see that I did not intentionally send?
- B3. What can third parties embedded in a website see?
 - B3a. What is a third-party script, and why do sites load them?
 - B3b. How do advertising networks see me across multiple websites?
 - B3c. What is a tracking pixel, and how is it different from a cookie?
- B4. What can my router or Wi-Fi operator see?
 - B4a. What does HTTPS protect me from on public Wi-Fi?
 - B4b. What can a Wi-Fi administrator see that HTTPS does not hide?
- B5. What does my browser itself know, and does it share that?
 - B5a. What is browser telemetry?
 - B5b. How do browser-level accounts (Google, Firefox Sync) affect privacy?

I also took screenshots of some of the queries I typed and what search results I received, just for a visualization of what my search looked like:



tracking pixel vs cookie

Search Assist

Tracking pixels and cookies are both used to track user behavior online, but they function differently. Tracking pixels are tiny, invisible images embedded in web pages or emails that send data to servers when loaded, while cookies are small data files stored on a user's device that help identify users and personalize their experience.

Auto-generated based on listed sources. May contain inaccuracies. Was this helpful?

TechTarget
https://www.techtarget.com/whatis/feature/Tracking-pixel-vs-cookie-Whats-the-difference

Tracking pixel vs. cookie: What's the difference? - TechTarget

May 8, 2024 · Tracking pixels and cookies both track users while they browse the web. Learn the difference between the two here.

Usercentrics
https://usercentrics.com/knowledge-hub/pixel-cookie

Are Tracking Pixels and Cookies Legal? Differences Explained

Nov 14, 2025 · Learn about the differences between cookies and pixels, their uses, legality, and how to stay compliant with a range of privacy laws.

learnwebanalytics.com
https://learnwebanalytics.com/whats-the-difference-between-a-cookie-a-pixel-and-a-tag

What's the Difference Between a Cookie, a Pixel, and a Tag? - Lea...

Summary Cookies, pixels, and tags are all related, but there are key differences that separate the three elements. The biggest difference between a cookie and a pixel is who is trying to track you - a third party ad tool or the website itself.

What is A Cookie? What is A Pixel? A Real Life Example of A Cookie, A Pixel >

Before we get into cookies, you need to understand what a web server is. A server is a computer that delivers data to your computer. A web server delivers websites. An ad server delivers advertisements. Simple enough. So how do cookies relate to web servers? A cookie is a piece of code placed on your browser by a web server. Because you visited the...

See more on learnwebanalytics.com

Mailchimp
https://mailchimp.com/resources/pixel-tracking

Pixel Tracking vs Cookies: Key Differences Explained - Mailchimp

Discover the key differences between pixel tracking and cookies, their uses, advantages, and impact on privacy regulations.

CookieYes
https://www.cookieyes.com/knowledge-base/cookies-101/pixel-vs-cookie

What is the difference between a pixel and a cookie?

A pixel or tracking pixel is a transparent graphic embedded in a web page or an email to collect data from users. A cookie is a small piece of data stored on a user's device by the user's web browser when they visit a website.

RedTrack
https://www.redtrack.io/blog/tracking-pixel-vs-cookie

Tracking Pixel vs Cookie: 2026 Guide for Media Buyers

Nov 23, 2025 · Learn the key differences between tracking pixels and cookies in 2026, and how privacy updates and new tech reshape performance marketing strategies...

onadvertising.com
https://onadvertising.com/tracking-pixel-vs-cookies-whats-the-difference

Tracking Pixel Vs. Cookie: What's the Difference and Why It Matte...

How Tracking Pixels and Cookies Work Together In modern digital marketing, a tracking pixel and a cookie often work in tandem. A tracking pixel can trigger the setting or retrieval of a cookie to tag the user for retargeting purposes. For instance, when a user visits a product page, a tracking pixel can fire...

WebToffee
https://www.webtoffee.com/info-library/data-privacy/pixel-vs-cookie

Pixel vs Cookie: What's the Difference? - WebToffee

Feb 25, 2026 · Pixels are commonly used for ad tracking, conversion measurement, and email open rates. A cookie, on the other hand, is a small piece of data stored directly in the user's browser. Cookies track user preferences, login sessions, and behavior across visits.

Cookiebot
https://www.cookiebot.com/en/pixel-tracking

What Is Pixel Tracking? Definition & How It Works

Mar 16, 2026 · Learn what pixel tracking is, how tracking pixels work, and why they require consent under GDPR. Understand pixels vs cookies and privacy compliance requirements.

iubenda
https://www.iubenda.com/en/blog/tracking-pixel-vs-cookie-explained-and-why-it-should-mat...

Tracking pixel vs cookie explained (and why it should matter ... - i...

Tracking pixel vs cookie. These two technologies are both used for similar purposes, such as tracking behaviors or showing ads. However, a few things differentiate them. One thing is sure, though, is that you need to comply with data privacy laws when using them!

how do ad networks decide what ad to show me

Search Assist

Ad networks decide what ads to show you by using data collected from your browsing history, search queries, and profile information. This information helps them create targeted ads that are more likely to interest you.

Auto-generated based on listed sources. May contain inaccuracies. Was this helpful?

Facebook
https://facebook.com/help/44727887528796/?helpref=popular_articles

How Facebook ads use machine learning

Learn more about your ad choices. To decide which ads to show you, we use Advertiser audience selections and the results of our Ad auction to determine the best ad to show you at a given point in time. Our machine learning models are part of our ads delivery system that learns as it receives new...

Google Support
https://support.google.com/adsense/answer/97137hl=en

How ads are targeted to your site - Google AdSense Help

With placement targeting, advertisers choose specific ad placements, or subsections of publisher websites, on which to run their ads. Ads that are placement-targeted may not be precisely related to the content of a page, but are hand-picked by advertisers who've determined a match between what your...

advertising.com
https://advertising.com/ad-networks/how-ad-networks-work

How Ad Networks Work: A Complete Guide for Advertisers

Feb 5, 2026 · How do ad networks differ from ad exchanges? Ad networks collect ad inventory from multiple publishers and sell it to advertisers, acting as a middleman to simplify ad buying.

Information Technology and Innovation Foundation
https://itif.org/publications/2021/11/02/itif-technology-explainer-how-do-online-ads-work

ITIF Technology Explainer: How Do Online Ads Work?

Online ads support the vast array of free apps and content on the Internet. Advertisers can use ad networks to deliver relevant ads to users through two primary methods: Contextual Ads: Advertisers use ad networks to place ads based on the content or keywords on specific web pages. For example...

business.safety.google
https://business.safety.google/privacy/display-advertising-us

How Google uses your information for display advertising on part...

When you use websites and apps (or similar online services like Connected TV), you may see image or video ads appear on the page you are visiting. This is called display advertising. These display ads keep many of these websites, apps and services free of charge.

save my leads.com
https://savemyleads.com/blog/other/how-does-facebook-know-what-ads-to-show-me

How Does Facebook Know What Ads to Show Me - SaveMyLeads

Jun 19, 2024 · In today's digital age, personalized advertising has become a cornerstone of online experiences. Have you ever wondered how Facebook seems to know exactly what products or services you're interested in? This article delves into the sophisticated algorithms and data collection methods...

illuminetmedia.com
https://www.illuminetmedia.com/post/how-facebook-ads-decide-what-ads-will-be-placed-and...

How Facebook Ads Decide What Ads Will Be Placed and When: ...

Sep 9, 2024 · In today's digital age, Facebook ads are a powerful tool for small businesses to reach new audiences, increase sales, and boost brand awareness. But how does Facebook determine what ads will be placed and when? Let's break down the process and offer some actionable insights for small...

Internet Understood
https://internetunderstood.com/how-does-facebook-know-what-ads-to-show-you

How Does Facebook Know What Ads To Show You? - Internet Un...

The short answer is that Facebook knows a lot about you, and allows advertisers to choose what kinds of people they show their ads to. The long answer is really interesting, and as a professional Facebook advertiser, I can tell you a lot about how advertising on Facebook really works and what that means for...

MonetizeMore
https://www.monetizemore.com/blog/choose-ads-show-website

How 7 figure publishers pick ads to show on their sites?

How do you know? If you explore your AdSense or Ad Exchange dashboard, you'll see a tab that says "Allow and Block Ads." This is where you choose what ads to show on your site. You can select from a long list of ad categories to control which of these ads you intend to show or block from your website.

Facebook
https://www.facebook.com/help/20721634931757/?query=adverts

How does Facebook decide which ads to show me?

Update your Ads based on activity information from ad partners setting. You can choose whether we use information based on your activity on websites or apps off Meta technologies for the purpose of showing you ads in your ad settings.